



Cloud Native Observability

Prometheus Basics

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This article explores the fundamentals of Prometheus, an open-source monitoring tool for collecting, visualizing, and alerting on metrics data.

In this lesson, we explore the fundamentals of Prometheus, an industry-leading open-source monitoring tool used for collecting, visualizing, and alerting on metrics data.

What is Prometheus?

Prometheus is a powerful open-source monitoring solution that scrapes metrics from targets via HTTP endpoints. It is widely used to gather critical system and application metrics, store them in a time series database, and enable complex querying using its native language, PromQL. Additionally, Prometheus features built-in alerting that allows you to define rules for notifying teams when specific thresholds are exceeded.



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Prometheus is an open-source monitoring tool that collects metrics data, and provide tools to visualize the collected data

In addition, Prometheus allows you to generate **alerts** when metrics reach a user specified threshold

Prometheus collects metrics by scraping targets who expose metrics through an HTTP endpoint

Scraped metrics are then stored in a time series database which can be queried using Prometheus' built-in query language PromQL

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Types of Metrics Monitored

Prometheus is versatile in monitoring a broad range of metrics, including:

System Metrics:

- CPU utilization

- Memory usage

- Disk space usage

- Service uptime

Application-Specific Metrics:

- Exception counts

- Latency measurements

- Number of pending requests

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So what kind of metrics can Prometheus Monitor?

- CPU/Memory Utilization
- Disk space
- Service Uptime
- Application specific data
 - Number of exceptions
 - Latency
 - Pending Requests

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Data Integration

Prometheus is built to seamlessly collect metrics from a diverse range of sources—not just applications and operating systems but also networking devices, databases, and other critical IT infrastructure components. Its extensive integration capabilities make it a favorite among IT professionals.

Prometheus is highly extensible, enabling custom metric collection tailored to your specific requirements.

Focused on Numeric Data

Prometheus specializes in capturing numeric time-series data. It is engineered for metric collection and analysis, meaning it is **not designed** to monitor events, logs, or traces.

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Prometheus is designed to monitor time-series data that is **numeric**

What type of data should Prometheus **not** monitor

- Events
- System logs
- Traces

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Background and Development

Originally developed with sponsorship from SoundCloud, Prometheus has evolved significantly since its inception. In 2016, it joined the Cloud Native Computing Foundation, further solidifying its role in the cloud-native ecosystem. The tool is primarily written in Go, contributing to its performance and scalability.

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For more detailed information and comprehensive documentation, visit the official Prometheus website at prometheus.io/docs.

Additional Resources

For further insights into Prometheus and modern monitoring practices, consider exploring the following resources:

Resource Type	Use Case	Example Command or Link
Monitoring Tool	Setting up system and application metrics	Prometheus Documentation
Cloud Native	Integration with cloud-native solutions	Cloud Native Computing Foundation
Open Source	Community-driven tools and contributions	GitHub - Prometheus

Enhance your observability strategy by leveraging the best practices in modern monitoring with Prometheus.



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